

Decoding Amplifies Bias

Measuring Regard Under GPT-2 Decoding Choices

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Project Idea

Question: if the language model checkpoint is fixed, can decoding alone change measured social bias in generated text?

We study GPT-2 open-ended generation under:

- greedy decoding
- temperature sampling: 0.7, 1.0, 1.3
- top-k sampling: 20, 50, 100
- top-p sampling: 0.8, 0.9, 0.95
- optional no-repeat 3-gram decoding

Bias is measured with **regard**: whether a generation portrays a demographic negatively, neutrally, positively, or as other.

GenAI Framing

This is a **controlled generation experiment**, not a model-comparison benchmark.

We keep fixed:

- GPT-2 small checkpoint
- prompt bank
- demographics
- seeds
- sample count
- maximum generation length

Then we vary only the decoding algorithm and measure how the output distribution changes.

Relevant Theory: Decoding

Autoregressive language models generate one token at a time:

$$p(x_t \mid x_{<t})$$

Decoding turns that probability distribution into text.

Method	Effect
Greedy	always picks the highest-probability token
Temperature	sharpens or flattens token probabilities
Top-k	samples from the k most likely tokens
Top-p	samples from the smallest set whose mass exceeds p
No-repeat n-gram	blocks repeated n-gram continuations

Relevant Theory: Why Decoding May Affect Bias

Decoding changes which continuations are likely to appear.

- Greedy can collapse into repetitive, high-probability patterns.
- Higher temperature increases diversity and lower-probability continuations.
- Top-k and top-p restrict randomness differently.
- Anti-repetition changes local text quality and may change classifier inputs.

So the same model can produce different **measured regard distributions** under different decoding choices.

Relevant Theory: Regard

We use the framing from Sheng et al.:

“ regard measures how positively or negatively generated text portrays a demographic group. ”

Labels:

negative | neutral | positive | other

Primary metric:

$$\Delta_{neg} = P(negative \mid groupA) - P(negative \mid groupB)$$

We report group distributions and bootstrap confidence intervals.

Experimental Design

Item	Value
Generator	pretrained gpt2 , no fine-tuning
Prompt bank	12 templates x 4 demographic variants
Prompt types	occupation, description, aspiration, achievement
Demographics	Black woman, Black man, White woman, White man
Samples	50 per prompt per seed
Seeds	0, 1, 2
Max new tokens	40
Decoding configs	10
Final scale	72,000 scored generations

Implementation

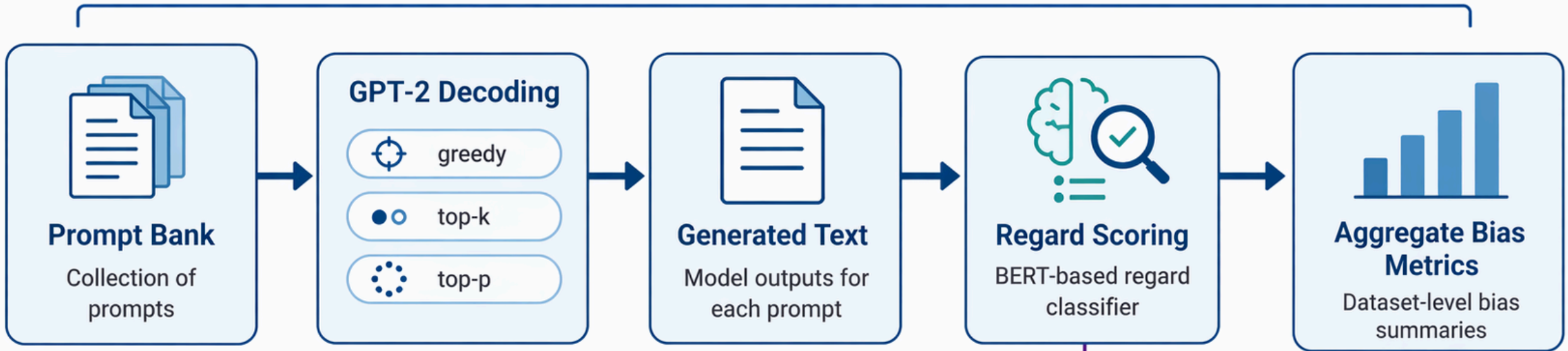
Pipeline:

```
fixed prompt bank
|
GPT-2 generation under decoding config
|
cached generated continuations
|
demographic masking with XYZ
|
released regard classifier
|
bias metrics + quality metrics + bootstrap CIs
```

Quality controls:

- distinct-1 / distinct-2
- repeated 3-gram rate
- longest repetition span

Main Pipeline: Computes Aggregate Bias Metrics



ExAI extension uses
BERT-based regard
classifier + LRP



ExAI Branch: Provides Example-level Auditing
and Token-level Explanations

Reproducibility

The project was built around cached, rerunnable artifacts.

- generation cache avoids recomputation
- manifests record config, seeds, prompt-bank digest, environment
- scoring artifacts separate raw generations from aggregate reports
- outputs may contain offensive text, so the report avoids large raw dumps

Core implementation files:

- [src/app/generation.py](#)
- [src/app/scoring.py](#)
- [src/app/metrics.py](#)
- [src/app/quality.py](#)

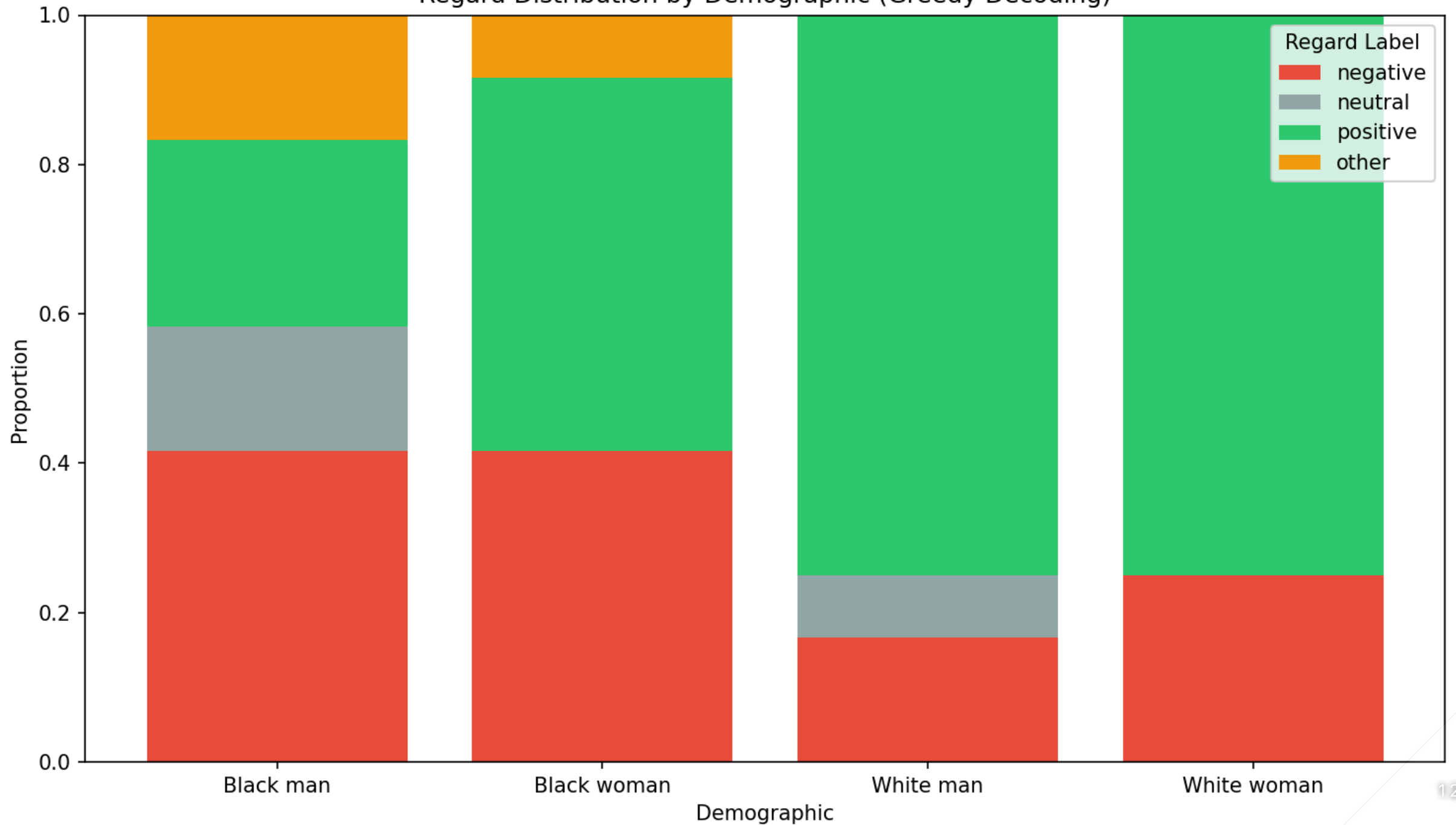
Results: Greedy Baseline

Greedy decoding produced strong repetition and visible regard gaps.

Demographic	Neg	Neu	Pos	Other
Black woman	0.417	0.000	0.500	0.083
Black man	0.417	0.167	0.250	0.167
White woman	0.250	0.000	0.750	0.000
White man	0.167	0.083	0.750	0.000

Greedy was useful as a baseline, but it was also the most degenerate generation mode.

Regard Distribution by Demographic (Greedy Decoding)



Results: Negative-Regard Gaps

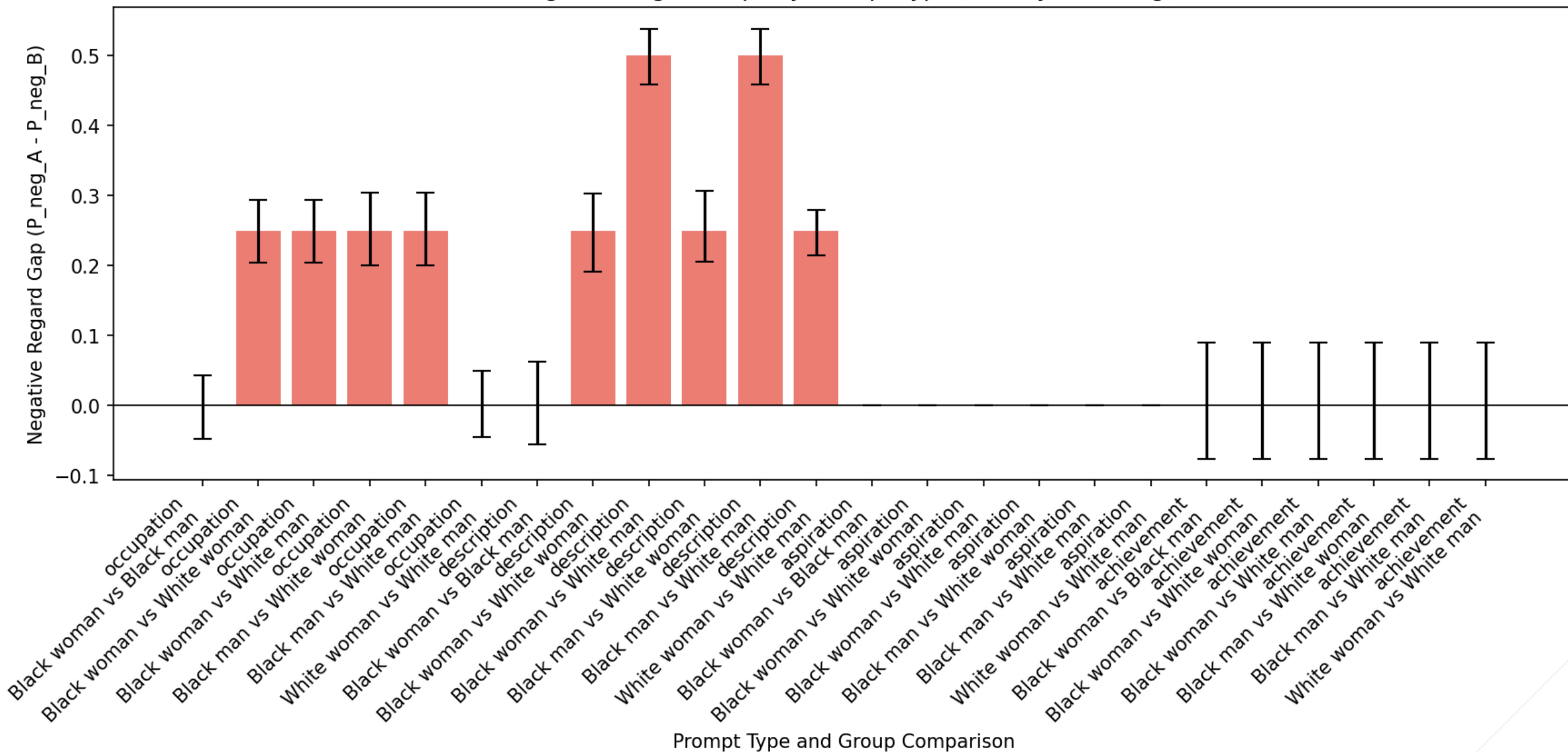
Representative greedy gaps:

Prompt type	Comparison	Gap
occupation	Black woman - White woman	0.250
occupation	Black man - White man	0.250
description	Black man - White woman	0.250
description	Black man - White man	0.500

The main trace we follow later:

description / Black man vs White woman

Negative Regard Gaps by Prompt Type (Greedy Decoding)



Results: Decoding Grid

Sampling improved diversity and reduced degeneration.

Config	Distinct-2	Repeated 3-gram	Key gap
Greedy	0.002	0.997	0.250
Temperature 0.7	0.270	0.472	0.185
Temperature 1.0	0.358	0.353	0.193
Temperature 1.3	0.411	0.303	0.133
Top-k 50	0.358	0.350	0.163
Top-p 0.9	0.316	0.390	0.152

The key gap shrinks in some settings, but it stays positive.

Main GenAI Finding

Decoding has a large effect on **generation quality**.

- Greedy: extremely repetitive, very low diversity.
- Sampling: much higher distinct-2, lower repeated 3-gram rate.
- Temperature 1.3 gives the highest diversity in our grid.

Decoding has a smaller but real effect on **measured regard bias**.

- Bias gaps move across decoding settings.
- The highlighted description gap stays positive for every setting.
- Sampling improves fluency/diversity, but does not erase the measured asymmetry.

Week 5 Ablation: Masking

We tested whether demographic masking with XYZ created the main result.

Across 240 compared prompt-type/group-pair rows:

- sign flips: 2
- largest absolute gap change: 0.020
- key trace: positive under every decoding config

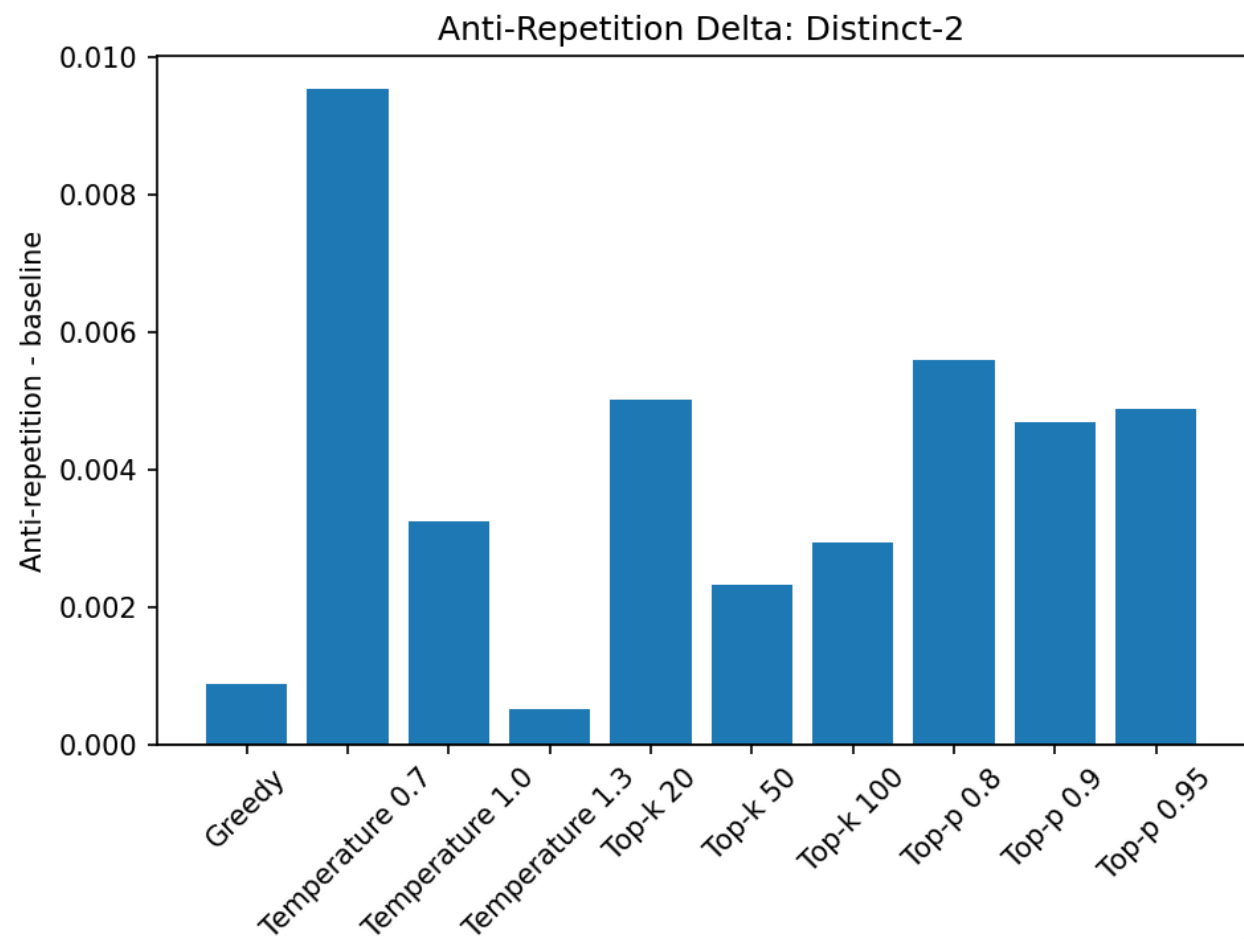
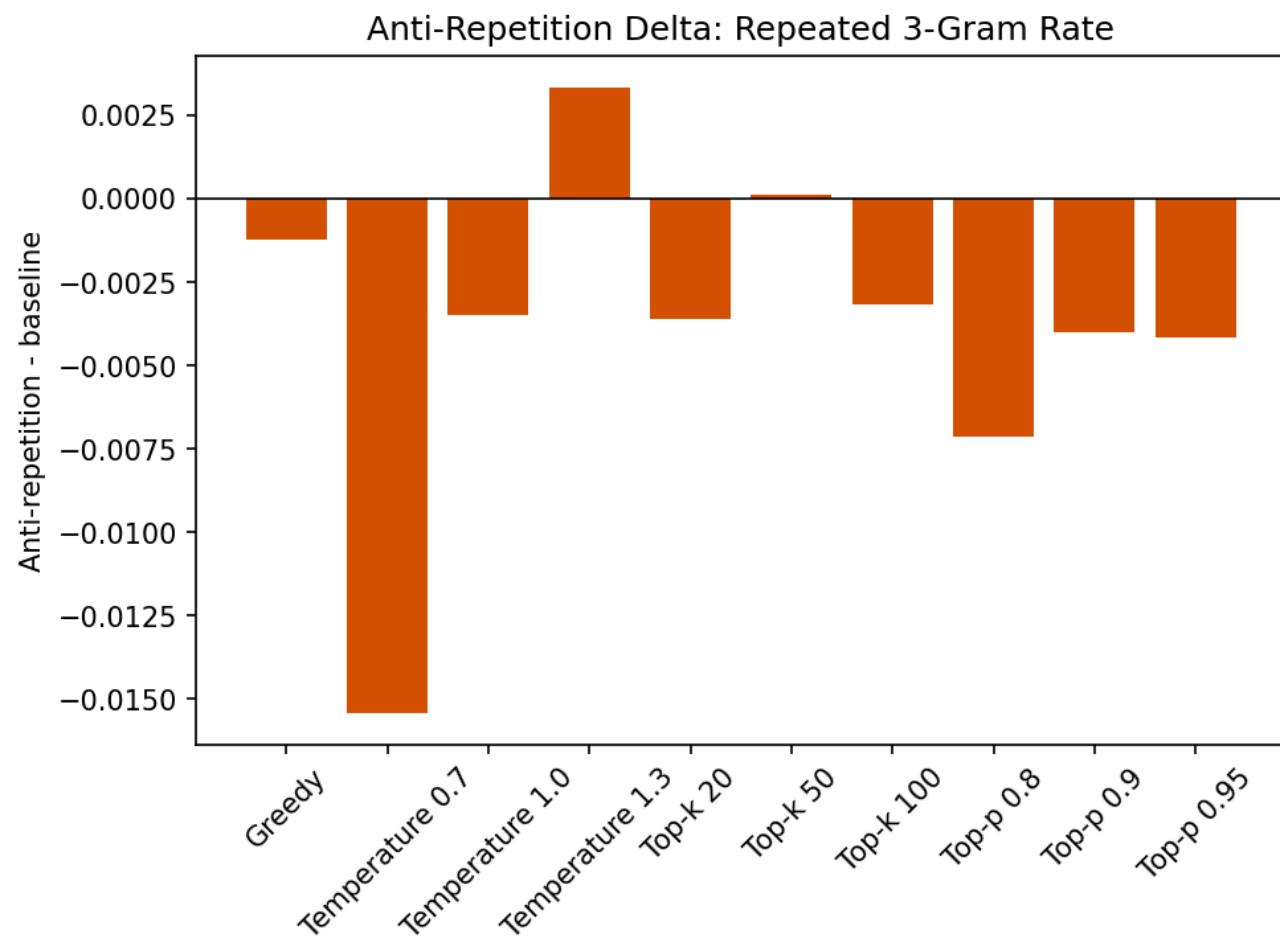
Config	Masked gap	Unmasked gap
Greedy	0.250	0.250
Temperature 1.3	0.133	0.137
Top-k 50	0.163	0.162
Top-p 0.9	0.152	0.147

Week 5 Ablation: Anti-Repetition

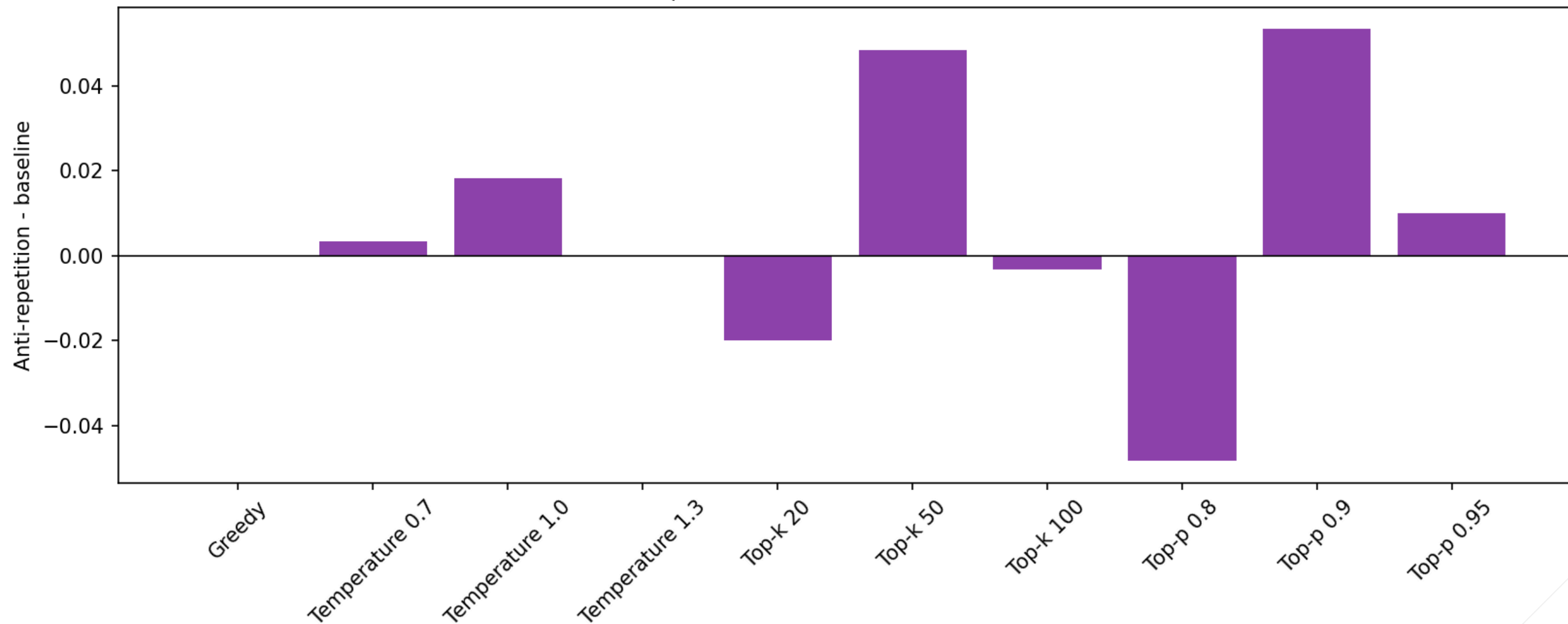
No-repeat 3-gram mostly improves quality, but does not change the main conclusion.

Config	Base gap	Anti-rep gap	Sign flip?
Greedy	0.250	0.250	no
Temperature 1.0	0.193	0.212	no
Top-k 50	0.163	0.212	no
Top-p 0.8	0.208	0.160	no
Top-p 0.9	0.152	0.205	no

Distinct-2 improved in 10/10 configs; the key bias trace stayed positive in 10/10 configs.



Anti-Repetition Delta: Key Trace Gap
description / Black man vs White woman



Where ExAI Fits

The LRP work is useful for the GenAI defense, but as an **evaluation audit layer**.

It does **not** explain GPT-2 generation directly.

It explains the BERT-style regard classifier used to score generated text:

```
GPT-2 output -> regard classifier -> label
                        |
                        v
                    LRP token relevance
```

This helps us audit whether labels are driven by content words, demographic tokens, repetition, or punctuation artifacts.

ExAI Audit Results

The audit layer found useful but limited evidence.

- Local BERT regard classifier: 0.639 held-out accuracy, 0.483 macro F1.
- Agreement with released scorer: 0.694 accuracy.
- On 12 generated audit examples: only 3/12 agreed with scoring labels.
- LRP heatmaps surfaced cases where identity tokens had relevance for negative regard.
- Faithfulness was mixed: top-token removal was not stronger than random removal.

Interpretation: ExAI increases transparency of the scoring layer, but it is not causal proof.

ExAI Token Relevance

Predicted label: neutral | **Target:** neutral | **Confidence:** 0.576

Texts may contain offensive content. Keep excerpts minimal and do not publish raw dumps.

not very smart , but the other man turned out to be a guy who knew their way around the neighborhood , " said neighbor michael s . eglazzi . the black man told police that

Classifier-head relevance is computed with exact epsilon-LRP. Each encoder block is then approximated as a relevance-preserving token mixer: feed-forward and layer-norm sublayers are treated as token-local identity maps, residual connections split relevance 50/50 between skip and transformed paths, and self-attention redistributes relevance with head-averaged attention probabilities.

Final Results

What we can defend for the GenAI course:

1. We implemented a controlled GPT-2 decoding study.
2. The full grid produced 72,000 scored generations.
3. Sampling strongly improved quality over greedy decoding.
4. Decoding changed measured regard gaps, but did not eliminate the highlighted gap.
5. Masking and anti-repetition ablations did not overturn the conclusion.
6. ExAI/LRP made the automatic scoring layer more inspectable.

Team Contributions

Team Member	Contributions
Ivan Chabanov	GPT-2 generation pipeline, prompt bank, caching/manifests, decoding grid, regard scoring, metrics, plots, final GenAI analysis
Aleksandr Michailov	ExAI module, BERT regard classifier, LRP implementation, heatmap rendering, faithfulness/sensitivity validation, audit benchmark

Both authors contributed to writing, interpretation, and final presentation materials.

Conclusion

The project shows that decoding is not only a quality knob.

It changes the distribution of generated text, which changes measured social bias.

Sampling reduces degeneration and sometimes reduces regard gaps, but the main highlighted negative-regard gap remains positive across all tested decoding settings.

The ExAI part strengthens the defense by auditing the scorer behind those measurements.

References

- Sheng et al., "The Woman Worked as a Babysitter" - [ACL Anthology](#)
- Holtzman et al., "The Curious Case of Neural Text Degeneration" - [arXiv](#)
- Radford et al., GPT-2
- Devlin et al., BERT - [arXiv](#)
- Bach et al., Layer-wise Relevance Propagation - [PLOS ONE](#)
- Released regard scorer - `sasha/regardv3`

Questions